12.3.2 From Definitions to an Algorithm

Moving from the definitions in the previous section to an algorithm is surprisingly simple.

For a given point `p`, we need to check how many of its neighbors lie within a radius ε; if there are more than a certain number `m`, then we mark `p` as a core point, and add its neighbors to the same cluster. Otherwise: never mind, we do nothing. Figure 12.10 illustrates this step, that is going to be repeated for every point in the dataset, with the help of a set to keep track of the points that we want to process next (in any order) for current cluster.

Now we have to ask ourselves, what happens when we process a point `w` that is not a core point, but is directly reachable from a core point `p`? Is it fine if we don’t take any action while processing `w`?